

Ponni Selvakumara Chinnaiyan

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EDUCATION

- **Kongu Engineering College** Erode , India
Master of Engineering in Computer Science; GPA: 8.44/10.0 Aug. 2015 – May. 2017
- **PSG College of Technology** Coimbatore, India
Bachelor of Engineering in Computer Science; GPA: 7.48/10.0 Jul. 2011 – May. 2015

SKILLS

- **Backend:** Python, Java, C++, C; **Frontend:** Javascript, HTML, CSS;
- **Database:** MySQL, MS Access, Elastic search, Oracle DB; **Test Frameworks :** JUnit, GoMockery
- **Others:** Android, Git, Svn, AWS, OCI, Docker, kubernetes
- **Related Coursework:** Distributed Systems, Data Science, Machine Learning, Wireless Sensor Networks, Network Security, Software Engineering, Operating System, Data structures, Algorithms

EXPERIENCE

- **Velalar College of Engineering and Technology** Erode, India
Assistant Professor Sep 2018 - Jul 2022
 - Academic professional with 4 years' experience in higher education, specializing in programming and curriculum design.
 - Instructed over **10+ core subjects**, including Java, Python, C, and C++, for **600+** undergraduate students.
 - Supervised **30+ student projects**, resulting in **5+ journal publications**
 - Held key academic roles: Class Advisor, Timetable In-Charge, Association & CSI Coordinator, and Newsletter Editor
 - Led syllabus design for multiple subjects, enhancing course alignment with industry trends
- **Kongu Engineering College** Erode, India
Research Assistant under Prof. R.R.Rajalaxmi Jun 2017 - Aug 2018
 - Worked for the research project approved by **Indian Council of Social Science Research(ICSSR)**
 - Project Title: A Study and Analysis of the Usage of Web and Social Media Networks to Predict the Behavioral Aspects and Learning Experience of Students in Engineering Education using Data Mining
 - Collected and analyzed behavioral and academic data from over **1,500 engineering students** across **10+ institutions**
 - Developed multiple regression models predicting student performance from behavioral and social media data, with prediction errors consistently within ± 0.5 of actual scores.
 - Led end-to-end design and implementation, covering requirements gathering, data preprocessing, feature engineering, model training, and evaluation.
 - Journal Publication: [Regression Model for Predicting Engineering Students Academic Performance](#) - Citation Index: 23

SELECTED PROJECTS

- **Novel Clustering Protocol for Large Scale Wireless Sensor Networks:** Designed and developed an efficient protocol for rapid data transmission from source to sink nodes, utilizing a radial-shaped clustering method combined with greedy and chain techniques. Achieved a **4x increase in packet collection rate compared to existing systems**.
Journal Publications: [Multi Objective Clustering Techniques with guaranteed delivery in Wireless Sensor Networks](#)
- **Medical Image Classification:** Designed and implemented a decision tree algorithm combined with association rule mining to classify and detect abnormalities in CT scan images, achieving an accuracy of 81% and precision of 89% on test dataset.

SELECTED JOURNAL PUBLICATIONS

- Regression Model for Predicting Engineering Students Academic Performance - International Journal of Recent Technology and Engineering (IJRTE) - ResearchGate - April 2019
- Student Vehicle Management - South Asian journal of Engineering and Technology - April 2019
- Student Monitoring System - South Asian journal of Engineering and Technology - April 2019
- Flower Pollination Based Task Scheduling With Cost Minimization Effort Using Multi Cloud Resource Provisioning - International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET) - April 2020
- Trusted Cloud based Storage and Retrieval Process using Symmetric Searchable Encryption - International Journal of Engineering Research and Technology (IJERT) – August 2020
- Enhanced Crow Search Algorithm for Task Scheduling in Cloud Computing - IOP Conference Series: Materials Science and Engineering – ResearchGate - February 2021
- Two Level QRFIE Code for Private Message Sharing - International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET) – May 2021

PATENT

- **IPR Patent:** System for Doctor Recommendation and Availability Tracking in Pandemic Scenarios Filed under Intellectual Property Rights (IPR) – **Application No.: 2020041057559**, Filed on: [31 Dec, 2020] Co-inventor of a system that leverages doctor ratings and real-time availability data to recommend suitable healthcare providers, enhancing responsiveness during pandemic or high-demand situations. Designed to support effective medical resource management using dynamic status tracking and patient feedback integration.

ACCOMPLISHMENTS

- Completed the **Effective Engineering Teaching** in Practice course on **NPTEL**, earning an **Elite** certificate with a score of **90%** in March 2021.
- Completed the **Python** Programming course through the **INFYQTQ platform by Infosys Limited**, earning an **A+ grade** in May 2020.
- Attended **Train The Trainer Program on Java Programming** conducted by **Infosys Limited** in June, 2021
- Completed self-paced courses including **MATLAB Onramp, Machine Learning Onramp, Deep Learning Onramp, Cisco Networking Academy® Cybersecurity Essentials, Cisco Networking Academy® Introduction to Packet Tracer, and AWS Cloud** Literacy Inventor, achieving strong scores across all programs.
- Participated in multiple conferences and workshops within the data science domain to stay current with industry trends and enhance technical expertise.